

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2018-2019)

PRE-MEDICAL : NURTURE PHASE-III

TEST # 04

DATE : 09-09-2018

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **180**.
5. **Each correct answer carry 1 mark. While one third mark will be deducted for each incorrect answer.**
6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet.**

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MINOR TEST # 04	Topics	DATE : 09 - 09 - 2018
PRE-MEDICAL NURTURE COURSE : PHASE - III (PATTERN : NEET)		
PHYSICS	: Kinematics ,NLM	
CHEMISTRY	: Periodic Table ,Chemical Bonding	
BIOLOGY	: The Living World , Biological Classification, Animal Kingdom (Introduction To coelenterata)	

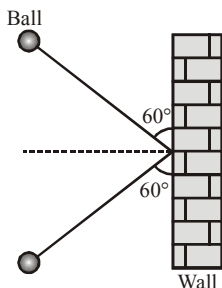
Next Up Coming Test

MINOR TEST # 05	Topics	DATE : 30 - 09 - 2018
PRE-MEDICAL NURTURE COURSE : PHASE - III (PATTERN : NEET)		
PHYSICS	: Motion in 2D, NLM, Friction	
CHEMISTRY	: Atomic Structure, Periodic Table, Chemical Bonding, Gaseous State	
BIOLOGY	: Monera & Protista Kingdom, Animal Kingdom (Ctenophora to Mollusca)	

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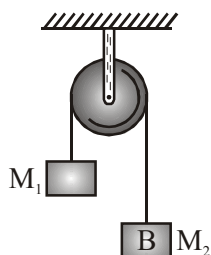
BEWARE OF NEGATIVE MARKING

1. A ball of mass 3 kg, moving with a speed of 100 m/s, strikes a wall at an angle 60° (as shown in figure). The ball rebounds at the same speed and remains in contact with the wall for 0.2 sec; the force exerted by the ball on the wall is :-



- (1) $1500\sqrt{3}$ N (2) 1500 N
(3) $300\sqrt{3}$ N (4) 300 N

2. Two masses $M_1 = 5$ kg and $M_2 = 10$ kg are connected at the ends of an inextensible string passing over a frictionless pulley as shown. When the masses are released, then the acceleration of the masses will be :-

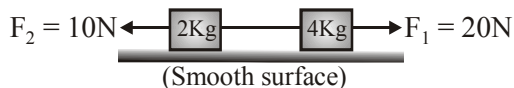


- (1) g (2) $g/2$ (3) $g/3$ (4) $g/4$

3. A force of 2 kgf is applied at one end of a spring balance kept horizontally and an equal force of 2 kgf is applied at the other end in the opposite direction, simultaneously. Then the reading on the spring balance is :-

- (1) 2 kgf (2) 4 kgf
(3) 0 kgf (4) 1 kgf

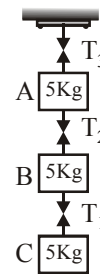
4. The tension in the string connected between blocks is :-



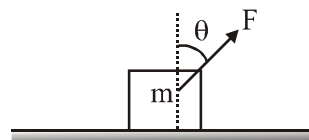
- (1) $\frac{80}{3}$ N (2) $\frac{40}{3}$ N
(3) $\frac{50}{7}$ N (4) 26 N

5. The value of $\frac{T_3}{T_1}$ is :-

- (1) 1
(2) 2
(3) 3
(4) $\frac{3}{2}$

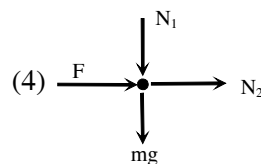
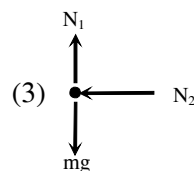
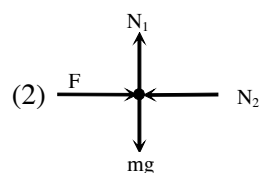
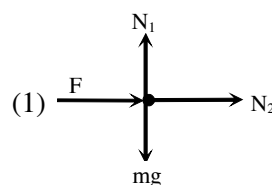
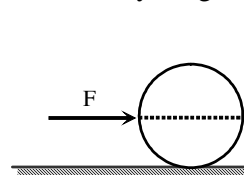


6. In the given arrangement, the normal force applied by block on the ground is :-



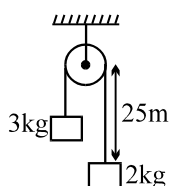
- (1) mg (2) $mg - F \cos \theta$
(3) $mg + F \cos \theta$ (4) $F \cos \theta$

7. A ball of mass m kept at the corner as shown in the figure, is acted by a horizontal force F. The correct free body diagram of ball is

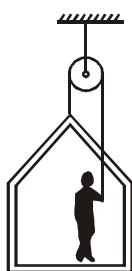


8. Find the time in which 2 kg mass will strike the pulley after being released from rest.

($g = 10 \text{ m/s}^2$)



- (1) 10 sec (2) 2 sec
(3) 5 sec (4) 4 sec
9. A man of mass m stands on a frame of mass M . He pulls on a light rope, which passes over a pulley. The other end of the rope is attached to the frame. For the system to be in equilibrium. What force must the man exert on the rope ?

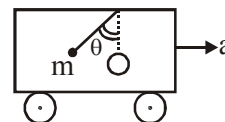


- (1) $\frac{(M+m)g}{2}$ (2) $(M+m)g$
(3) $(M-m)g$ (4) $(M+2m)g$
10. A ball is released from the top of a tower of height h meters. It takes T seconds to reach the ground. What is the position of the ball at $T/3$ second ?
- (1) $h/9$ meters from the ground
(2) $7h/9$ meters from the ground
(3) $8h/9$ meters from the ground
(4) $17h/18$ meters from the ground
11. If $\vec{\text{velocity}} \cdot \vec{\text{acceleration}} < 0$ at an instant, then speed of particle at this instant is :
- (1) increasing
(2) Decreasing
(3) Neither increasing, nor decreasing
(4) More information is needed
12. A particle is projected vertically upwards in the presence of uniform gravity ($g=10\text{m/s}^2$). Its total time of flight is more than 10 second. The distance travelled by particle in last second of its upward journey is :
- (1) 10 m (2) 5 m
(3) 2.5 m (4) 1.25 m

13. A projectile is projected on a horizontal ground. It reaches a particular height at two instants t_1 and t_2 . Then time of flight of projectile is

- (1) $\sqrt{t_1 t_2}$ (2) $t_1 t_2$
(3) $\frac{1}{2}(t_1 + t_2)$ (4) $t_1 + t_2$

14. If trolley accelerates horizontally with acceleration a then bob is displaced backward from its initial vertical position. The angular deflection of the bob in equilibrium is :-

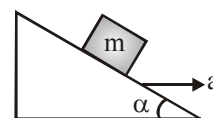


- (1) $\theta = \cos^{-1} \left(\frac{a}{g} \right)$ (2) $\theta = \sin^{-1} \left(\frac{a}{g} \right)$
(3) $\theta = \cot^{-1} \left(\frac{a}{g} \right)$ (4) $\theta = \tan^{-1} \left(\frac{a}{g} \right)$

15. The ratio of the weight of a man in a stationary lift and in a lift accelerate downwards with a uniform acceleration ' a ' is 3:2 The acceleration of the lift is :-

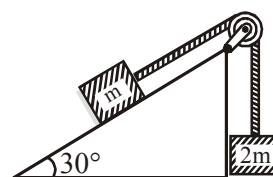
- (1) $g/3$ (2) $g/2$ (3) g (4) $2g$

16. A block is kept on a frictionless inclined surface with angle of inclination α . The incline is given acceleration ' a ' to keep the block stationary. Then a is equal to :-



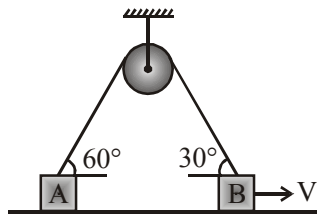
- (1) $\frac{g}{\tan \alpha}$ (2) $g \operatorname{cosec} \alpha$
(3) g (4) $g \tan \alpha$

17. Two blocks of masses m and $2m$ are connected by a light string passing over a friction less pulley, as shown in figure. If the system is released, the blocks move with an acceleration equal to :-



- (1) $\frac{g}{4}$ (2) $\frac{g}{3}$ (3) $\frac{g}{2}$ (4) g

18. For the given diagram when block B is pulled with velocity V then velocity of block A will be :-

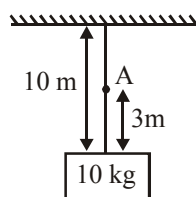


- (1) V (2) $\frac{V}{\sqrt{3}}$
 (3) $V\sqrt{3}$ (4) $\frac{\sqrt{3}}{2}V$

19. A car is moving in east direction and retarding. Two persons P and Q are sitting in same car. When P observes Q he will :-

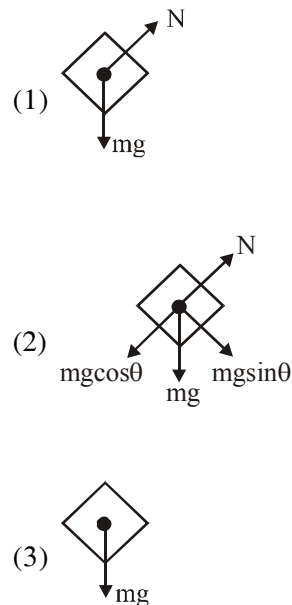
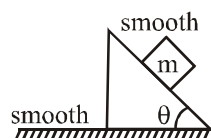
- (1) Apply a pseudo force in east direction
 (2) Apply a pseudo force in west direction
 (3) not apply any pseudo force
 (4) apply a pseudo force in any direction.

20. The adjoining figure shows a block of mass 10 kg connected to free end of a rope of mass 10 kg and length of 10 m. The tension of the rope at point A is ($g = 10 \text{ m/s}^2$):-



- (1) 170 N (2) 30 N
 (3) 130 N (4) 70 N

21. The free body diagram of the block of mass m is:-



- (4) None of the given

22. A rocket consumes fuel at the rate of 100 kg/s. The exhaust gases are ejected at a speed of $5 \times 10^4 \text{ m/s}$. Neglecting the effect of gravity, the force experienced by the rocket is :-

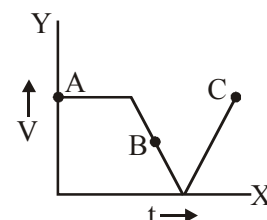
- (1) $5 \times 10^2 \text{ N}$ (2) $5 \times 10^4 \text{ N}$
 (3) $5 \times 10^6 \text{ N}$ (4) 50 N

23. In the two cases respectively, the ratio of contact force between the blocks will be :-



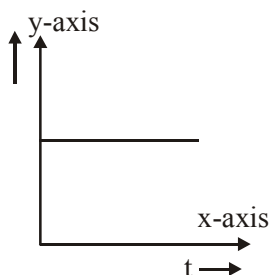
- (1) 1 : 4 (2) 2 : 1
 (3) 1 : 1 (4) 2 : 3

24. The velocity time graph of a body is shown in Figure. It implies that at point B :-

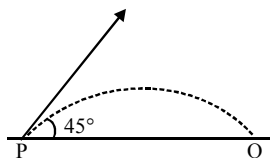


- (1) The acceleration is zero
 (2) There is a acceleration towards motion
 (3) There is a acceleration which opposes motion
 (4) None of the above

25. In the graph shown in figure, which quantity associated with projectile motion is plotted along y-axis



- (1) Range (2) Height
(3) Horizontal velocity (4) Vertical velocity
26. A projectile of mass m is fired with velocity v from the point P at an angle 45° with the horizon. The magnitude of change in momentum when it passes through the point Q on the same horizontal line on which P lies is :-

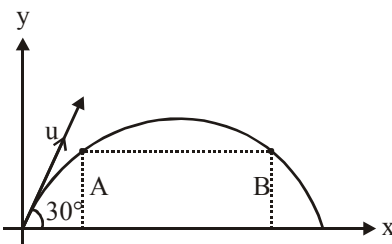


- (1) $mv\sqrt{2}$ (2) $\frac{1}{2}$ (3) Zero (4) mv
27. A particle is thrown such that its time of flight is 10s and horizontal range is 500m.
(i) $H_{\max} = 125\text{m}$
(ii) $u = 50\sqrt{2}\text{m/s}$
(iii) $\theta = 45^\circ$
(iv) velocity at highest point = 0 m/s
(1) (i), (ii), (iii) are correct
(2) (ii), (iii), (iv) are correct
(3) (i), (ii), (iv) are correct
(4) all options are correct
28. A particle having initial velocity 4 m/s moves with a constant acceleration 1 m/s^2 for a time interval of 10 seconds in straight line. Find the total distance travelled in 10 seconds :-
(1) 90 m (2) 10 m
(3) 80 m (4) 20 m
29. Two bodies A and B have masses M and m respectively, where $M > m$ and they are at a distance d apart. Equal force is applied to them so that they approach each other. The position where they hit each other is :-
(1) nearer to B
(2) nearer to A
(3) at equal distance from A and B
(4) cannot be decided

30. A ball is projected horizontally with a speed 40m/s in a vertical plane from the top of the tower of height 45m. The time taken to reach at the ground and speed of the ball just before it reaches at the ground will be :

- (1) 5 sec, 50 m/s (2) 3 sec, 50 m/s
(3) 2.5 sec, 25 m/s (4) 4 sec, 40 m/s

31. A particle is thrown with a velocity of $u\text{ m/s}$. it passes A and B as shown in figure at time $t_1 = 1\text{ s}$ and $t_2 = 3\text{ s}$. maximum height will be ($g = 10\text{ m/s}^2$) :-



- (1) 20 m (2) 10 m
(3) 40 m (4) 5 m

32. A particle starting from rest, moves with a uniform acceleration and covers x meters in the first 5 second. The same particle will cover the following distance in the next 5 seconds :-

- (1) x metre (2) $2x$ metre
(3) $3x$ metre (4) $4x$ metre

33. To a man standing, rain appears to fall vertically at 40m/s. When he walks with speed of 30m/s it appears to meet him at an angle x° with vertical and speed $v\text{ m/s}$. Find value of x & v :-

- (1) 37° , 40 (2) 37° , 50
(3) 53° , 30 (4) 53° , 50

34. The distance s meter covered by a body in t second is given by $s = 3t^2 - 8t + 5$, the body will stop after

- (1) 1s (2) $\frac{3}{4}\text{s}$ (3) $\frac{4}{3}\text{s}$ (4) 4s

35. The area under velocity- time graph gives :-

- (1) displacement
(2) force
(3) change in velocity
(4) velocity

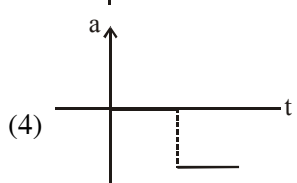
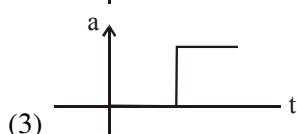
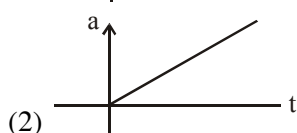
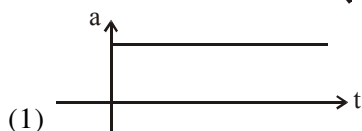
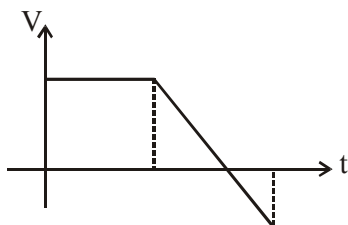
36. Four bodies P, Q, R and S are projected with equal speed having angles of projection 15° , 30° , 45° and 60° with horizontal respectively. The body having shortest range is :-

- (1) P (2) Q (3) R (4) S

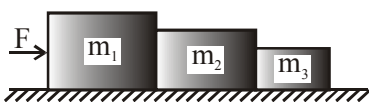
37. An object is thrown horizontally from a tower H meter high with velocity of $\sqrt{2gH}$ m/s. Its speed on striking the ground will be :-

(1) $\sqrt{2gH}$ (2) $\sqrt{6gH}$
(3) $\sqrt{4gH}$ (4) $\sqrt{8gH}$

38. A particle is moving along X-axis. Its velocity time graph is given, acceleration time curve is :-



39. Three blocks of masses m_1 , m_2 and m_3 kg are placed in contact with each other on a frictionless table. A force F is



applied on the heaviest mass m_1 ; the acceleration of m_3 will be:

(1) $\frac{F}{m_1}$ (2) $\frac{F}{m_1 + m_2}$
(3) $\frac{F}{m_2 + m_3}$ (4) $\frac{F}{m_1 + m_2 + m_3}$

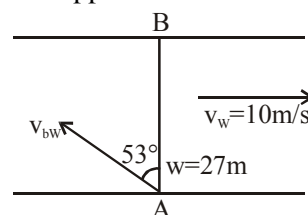
40. A cricket ball of mass 0.25 kg with speed 10 m s^{-1} collide with a bat and returns with same speed within 0.01 s. The force acted on bat is:

(1) 25N (2) 50N
(3) 250N (4) 500N

41. A large force is acting on a body for a short time. The impulse imparted is equal to the change in:

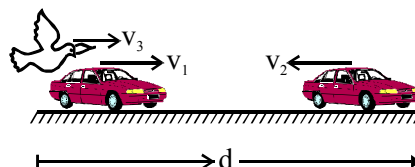
(1) acceleration (2) momentum
(3) energy (4) velocity

42. If $v_{bw} = 15 \text{ m/s}$, width of river is 27 m and flow of water is 10 m/s . Find drift of boat when it reaches on opposite bank of river (in metre).



(1) 3 (2) 1 (3) 6 (4) 4

43. A bird flies to and fro between two cars which move with velocities $v_1 = 20 \text{ m/s}$ and $v_2 = 30 \text{ m/s}$. If the speed of the bird is $v_3 = 10 \text{ m/s}$ and the initial distance of separation between them is $d = 2 \text{ km}$, Find the total distance covered by the bird till the cars meet:-

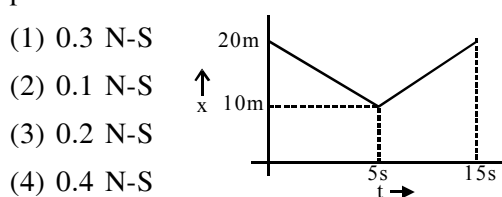


(1) 2000 m (2) 1000 m
(3) 400 m (4) 200 m

44. A person, reaches a point directly opposite on the other bank of a flowing river, while swimming at a speed of 5 m/s at an angle of 120° with the flow. The speed of the flow must be :-

(1) 2.5 m/s (2) 3 m/s
(3) 4 m/s (4) 1.5 m/s

45. Position-time graph for a particle of mass 100g is as shown in figure then impulse acting on the particle at $t = 5$ second is :-



(1) 0.3 N-S
(2) 0.1 N-S
(3) 0.2 N-S
(4) 0.4 N-S

46. Which of the following is the correct statements?
 (1) The second ionisation potential of an atom is always less than the first ionisation potential
 (2) The addition of an electron to a neutral atom is an endothermic process
 (3) Fluorine has the maximum electronegativity
 (4) The size of the cation is greater than the size of the neutral atom
47. The five successive ionisation energies of the element are 800, 2427, 3658, 25024 and 32824 kJ mol⁻¹ respectively. The number of valence electrons is :-
 (1) 3 (2) 5 (3) 4 (4) 2
48. Elements belonging to same group of the extended form of the periodic table have :-
 (1) same number of energy levels
 (2) same number of valence electrons
 (3) same number of total electrons
 (4) same ionisation enthalpy
49. When two atoms combine to form a molecule:-
 (1) energy is released
 (2) energy is absorbed
 (3) energy is neither released nor absorbed
 (4) energy may either released or absorbed
50. Which shows the highest lattice energy?
 (1) KCl (2) RbCl
 (3) CsCl (4) NaCl
51. Element X is strongly electropositive and Y is strongly electronegative. Both are univalent. The compound formed would be :-
 (1) X⁺ Y⁻ (2) X - Y
 (3) X⁻ Y⁺ (4) X → Y
52. In a triple bond there is sharing of :-
 (1) 3 electrons (2) 4 electrons
 (3) several electrons (4) 6 electrons
53. Which one of the following has pyramidal structure?
 (1) NH₃ (2) SiF₄
 (3) H₂O (4) BF₃
54. Which one of the following molecules are formed by p-p overlapping?
 (1) Cl₂ (2) HCl (3) H₂O (4) NH₃
55. The hybridization of S in SO₂ is :-
 (1) sp (2) sp² (3) sp³ (4) dsp²
56. The shape of a molecule which has 3σ bond pairs and two lone pair of electrons is :-
 (1) tetrahedral
 (2) T shape
 (3) trigonal bipyramidal
 (4) triangular planar
57. The types of bonds present in NH₄Cl are :-
 (1) electrovalent, covalent and coordinate
 (2) only ionic
 (3) only covalent
 (4) covalent and coordinate
58. In a regular octahedral molecule, MX₆, the number of X-M-X bonds at 180° is :-
 (1) 2 (2) 4 (3) 3 (4) 6
59. The strength of sigma bonds formed by axial overlap of s- or p-orbitals atoms decreases as :-
 (1) s-s > p-s > p-p
 (2) s-s > p-p > s-p
 (3) p-s > s-s > p-p
 (4) p-p > s-p > s-s
60. The linear structure is assumed by :
 A : SnCl₂ B : NCO⁻
 C : NO₂⁺ D : CS₂
 (1) A, B and C
 (2) B, C and D
 (3) A, C and D
 (4) B and D
61. Acetylene molecule contains :
 (Acetylene HC ≡ CH)
 (1) 5 sigma bonds
 (2) 4 sigma and 1 π-bond
 (3) 3 sigma and 2 π-bonds
 (4) 2 sigma and 3 π-bonds
62. A sp³ hybrid orbital contains :-
 (1) 1/4 s-character
 (2) 1/2 s-character
 (3) 2/3 s-character
 (4) 3/4 s-character

63. In which of the following the central atom does not use sp^3 hybrid orbitals in its bonding?
- (1) BeF_3^- (2) OH_3^+
(3) NH_2^- (4) NF_3
64. The ratio of sigma and pi-bonds in tetracyano ethylene is :-
-
- (1) 2 : 1 (2) 1 : 2
(3) 1 : 3 (4) 1 : 1
65. T-type shape is exhibited by the molecule :-
- (1) ClF_3 (2) $CHCl_3$
(3) CCl_4 (4) PCl_5
66. In XeF_2 , XeF_4 and XeF_6 the number of lone pairs on Xe is respectively :-
- (1) 2, 3, 1 (2) 1, 2, 3
(3) 4, 1, 2 (4) 3, 2, 1
67. In OF_2 , number of bond pairs and lone pairs of electrons are :-
- (1) 2, 6 (2) 2, 8
(3) 2, 2 (4) 1, 9
68. Which of the following is true?
- (1) Bond order $\propto \frac{1}{\text{Bond length}} \propto \text{bond energy}$
(2) Bond order $\propto \text{bond length} \propto \text{bond energy}$
(3) Bond order $\propto \frac{1}{\text{bond length}} \propto \frac{1}{\text{bond energy}}$
(4) None of these
69. Pair of species having identical shapes for molecules is :-
- (1) CF_4 , SF_4
(2) BF_3 , PCl_3
(3) XeF_2 , CO_2
(4) PF_5 , IF_5
70. Correct bond angle order is :-
- (1) $NH_3 > CH_4$
(2) $F_2O > H_2O$
(3) $Cl_2O > F_2O$
(4) $H_2O > NH_3$

71. Which of the following possess two lone pairs of electrons on the central atom and square planar in shape?
- (i) SF_4 (ii) XeO_4
(iii) XeF_4 (iv) ICl_4^-
- (1) I, III (2) III, IV
(3) II, IV (4) All
72. Molecular formulae and shapes of some molecules are given below. Choose the incorrect match :-
- | Formula | Shape |
|---------------|----------------------|
| (1) NH_3 – | Trigonal pyramidal |
| (2) SF_4 – | Tetrahedral |
| (3) ClF_3 – | T-shaped |
| (4) PCl_5 – | Trigonal bipyramidal |
73. Out of the following conversion which one results in the change of hybridization and geometry :-
- (1) NH_3 to NH_4^+
(2) H_2O to H_3O^+
(3) CH_4 to C_2H_6
(4) BF_3 to BF_4^-
74. The element of 4th period of the periodic table having maximum number of unpaired electrons in its ground state is :-
- (1) $_{20}Ca$ (2) $_{30}Zn$
(3) $_{24}Cr$ (4) $_{28}Ni$
75. d-block elements in long form of periodic table are placed :-
- (1) on the extreme left
(2) on the extreme right
(3) at the bottom
(4) in the middle
76. Diagonal relationship in periodic table is shown by elements of and periods.
- (1) I and II
(2) II and III
(3) III and IV
(4) I and III

- 77.** If we go from Li to F in the second period, there would be a decrease in :-
 (1) atomic mass
 (2) atomic radii
 (3) ionisation enthalpy
 (4) electronegativity
- 78.** Lowest ionisation enthalpy in periods is shown by :-
 (1) inert gases
 (2) halogens
 (3) alkali metals
 (4) alkaline earth metals
- 79.** In the sixth period of the extended form of periodic table, the orbitals are filled as :-
 (1) 6s, 5f, 6d, 6p
 (2) 5s, 5p, 5d, 6p
 (3) 6s, 6p, 6d, 6f
 (4) 6s, 4f, 5d, 6p
- 80.** Which of the following have no unit?
 (1) Electron gain enthalpy
 (2) Electronegativity
 (3) Ionisation enthalpy
 (4) Atomic radii
- 81.** Which one of the following belongs to representative group of elements in the periodic table?
 (1) Lanthanum (2) Argon
 (3) Chromium (4) Aluminium
- 82.** Polar molecule is :-
 (1) XeF₂ (2) XeF₄
 (3) XeO₃ (4) SF₆
- 83.** In the series carbon, nitrogen, oxygen and fluorine, electronegativity :-
 (1) decreases from carbon to fluorine
 (2) remains constant
 (3) decreases from carbon to oxygen and then increases
 (4) increases from carbon to fluorine
- 84.** Lanthanide contraction explains :-
 (1) density of the series
 (2) atomic volume of the series
 (3) chemical activity of the series
 (4) size of the atoms of the series
- 85.** Select the correct code for the following repulsive forces according to VSEPR theory :-
 (1) bond pair - bond pair > bond pair - lone pair > lone pair - lone pair
 (2) bond pair - lone pair > lone pair - lone pair > bond pair - bond pair
 (3) lone pair - lone pair > bond pair - bond pair > lone pair - bond pair
 (4) lone pair - lone pair > lone pair - bond pair > bond pair - bond pair
- 86.** Pentagonal bipyramidal structure contains bond angles approximately :-
 (1) 120°, 90°, 180° (2) 120°, 72°, 180°
 (3) 72°, 90°, 120° (4) 72°, 90°, 180°
- 87.** The hybridisation of orbitals of N atom in NO₃⁻, NO₂⁺ and NH₄⁺ are respectively :-
 (1) sp, sp², sp³ (2) sp², sp, sp³
 (3) sp, sp³, sp² (4) sp², sp³, sp
- 88.** Most favourable conditions for electrovalent (ionic) bonding are :-
 (1) low ionisation potential of one atom and high electron affinity of the other atom
 (2) high electron affinity and high ionisation potential of both the atoms
 (3) low electron affinity and low ionisation potential of both the atoms
 (4) high ionisation potential of one atom and low electron affinity of the other atom
- 89.** On hybridization of one s- and one p-orbitals, we get :-
 (1) two mutually perpendicular orbitals
 (2) two orbitals at 180°
 (3) four orbitals directed tetrahedrally
 (4) three orbitals in the plane
- 90.** Coordinate linkage is formed :-
 (1) by transfer of one electron from one atom to another
 (2) by the loss of one electron each from both the atoms
 (3) by sharing of one electron from each atom
 (4) when contribution of one electron pair is made by one atom and both the atoms share equally.

- | | |
|---|---|
| <p>91. The father of binomial nomenclature :-
 (1) Aristotle
 (2) Linnaeus
 (3) R.H. Whittaker
 (4) Carl Woese</p> <p>92. The position of bacteria in three kingdom classification proposed by Haeckel
 (1) Plantae (2) Animalia
 (3) Mychota (4) Protista</p> <p>93. Five kingdom system of classification was proposed by
 (1) Haeckel (2) Whittaker
 (3) Copeland (4) Linnaeus</p> <p>94. In five kingdom system, unicellular prokaryotes organism are grouped under the kingdom
 (1) Monera (2) Protista
 (3) Animalia (4) Fungi</p> <p>95. Two kingdom classification was proposed by :-
 (1) Whittaker (2) Haeckel
 (3) Linnaeus (4) Copeland</p> <p>96. Maintenance of internal favourable conditions despite changes in external environment is
 (1) Homeostasis (2) Enthalpy
 (3) Entropy (4) Steady state</p> <p>97. Which of the following features are shown by all organisms from prokaryotes to the complex Eukaryotes?
 (1) Extrinsic growth, Metabolism
 (2) Growth and reproduction
 (3) Metabolism and consciousness
 (4) Cellular organisation and self consciousness</p> <p>98. (A) The sum total of all chemical reactions taking place in our body is called metabolism.
 (B) In vitro metabolic reactions are living things.
 (1) Only (A) is correct
 (2) Only (B) is correct
 (3) Both (A) and (B) are incorrect
 (4) Both (A) and (B) are correct</p> | <p>99. Rattus rattus is a/an
 (1) Tautonym
 (2) Autonym
 (3) Homonym
 (4) Synonym</p> <p>100. Phylogeny is
 (1) Evolutionary history
 (2) Used in systematic arrangement of organisms
 (3) Based on the study of fossils
 (4) All of these</p> <p>101. Binomial nomenclature was first of all used in
 (1) Species plantarum
 (2) Philosophica botanica
 (3) Systema naturae
 (4) Hortus upplandicus</p> <p>102. Modern taxonomy includes study of
 (1) Internal structure and cell structure
 (2) Developmental process
 (3) Morphological characters
 (4) All of these</p> <p>103. Process of categorising different organisms on the basis of similarities and dissimilarities in their character is
 (1) Identification
 (2) Classification
 (3) Characterisation
 (4) Neosystematics</p> <p>104. In <i>Mangifera indica</i> L. The L represents
 (1) Language
 (2) Linnaeus
 (3) Lecto type
 (4) Sub species/variety</p> <p>105. ICBN stand for
 (1) International Code of Biological Names
 (2) International Code of Botanical Nomenclature
 (3) International Congress of Binomial Nomenclature
 (4) Indian Code of Binomial Names</p> |
|---|---|

- 106.** Founder of binomial nomenclature was
- (1) Darwin
 - (2) Mendel
 - (3) Linnaeus
 - (4) Lamarck
- 107.** 'Systema Naturae' was written by
- (1) Linnaeus
 - (2) Bauhin
 - (3) Lamarck
 - (4) Aristotle
- 108.** In taxonomy the first step is
- (1) Identification
 - (2) Nomenclature
 - (3) Classification
 - (4) Affinities
- 109.** In binomial nomenclature plants are given names in Latin because
- (1) Latin was the common Language
 - (2) Latin is a dead language
 - (3) Latin is a simple language
 - (4) All of these
- 110.** The zoological name *Naja naja*, the specific epithet is similar to the generic name. Thus it is an example of
- (1) Autonym
 - (2) Homonym
 - (3) Tautonym
 - (4) Synonyms
- 111.** Each category referred to as unit of classification infact represents a rank and is commonly referred as
- (1) Genus
 - (2) Species
 - (3) Taxon
 - (4) Tribe
- 112.** Which of the following taxonomic category is not amongst 7 obligate categories?
- (1) Division
 - (2) Kingdom
 - (3) Order
 - (4) Tribe
- 113.** Which of the following species concept is based upon sexual reproduction?
- (1) Biological concept
 - (2) Taxonomic concept
 - (3) Dynamic concept
 - (4) Static concept
- 114.** Biological concept of species was proposed by
- (1) Lamarck
 - (2) Mayr
 - (3) Linnaeus
 - (4) Aristotle
- 115.** Taxonomic studies of all known organisms have led to the development of hierarchy involving obligate categories.
- (1) 7
 - (2) 5
 - (3) 6
 - (4) 9
- 116.** Which of the following is represent a taxon?
- (1) Housefly
 - (2) Species
 - (3) Order
 - (4) Phylum
- 117.** Select incorrect match w.r.t. taxonomic categories of wheat.
- (1) Kingdom -Angiospermae
 - (2) Class - Monocotyledonae
 - (3) Order-Poales
 - (4) Family - Poaceae
- 118.** Anacardiaceae family belongs to the
- (1) Order poales and class dicotyledonae
 - (2) Order sapindales and kingdom spermatophyta
 - (3) Order sapindales and division angiospermae
 - (4) Order polymoniales and class dicotyledonae
- 119.** *Homo sapiens* belongs to class mammalia of phylum chordata and belong to order
- (1) Primata
 - (2) Diptera
 - (3) Carnivora
 - (4) Lapidoptera
- 120.** Interbreeding is possible between two members of same
- (1) Genus
 - (2) Species
 - (3) Order
 - (4) Family
- 121.** As we go from kingdom to species in a taxonomic hierarchy, the number of common characters
- (1) Will increase
 - (2) Will decrease
 - (3) Remain same
 - (4) May increase or decrease

- 122.** Which of the following taxon belong to lowest category?
- (1) Spermatophyta
 - (2) Monocotyledonae
 - (3) Plantae
 - (4) Wheat
- 123.** Term species was coined by
- (1) Linnaeus
 - (2) John Ray
 - (3) Aristotle
 - (4) Mayr
- 124.** The 'Mammals' taxonomically represent a
- (1) Family
 - (2) Class
 - (3) Order
 - (4) Phylum
- 125.** Which of the following categories includes all the others?
- (1) Order
 - (2) Phylum
 - (3) Kingdom
 - (4) Family
- 126.** Select a incorrect match.
- | Rank | Suffix |
|--------------|---------------|
| (1) Division | Phyta |
| (2) Class | Opsida |
| (3) Order | Ales |
| (4) Family | Phytina |
- 127.** Which of the following taxonomic categories are common for triticum and solanum?
- (1) Order, Class
 - (2) Class, Division
 - (3) Family, Order
 - (4) Division, Kingdom
- 128.** Which of the following taxonomic aids is based on the pair of contrasting characters?
- (1) Flora
 - (2) Key
 - (3) Herbarium
 - (4) Manuals

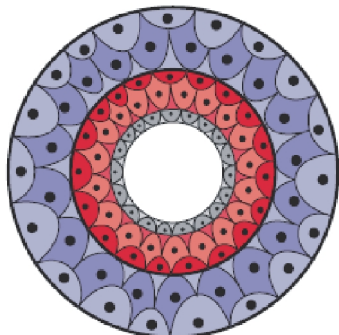
- 129.** Museum have collections of
- (1) Conserved animal species
 - (2) Conserved wild life
 - (3) Preserved plant and animal specimens
 - (4) Preserved animal specimen only
- 130.** Largest botanical garden in the world is
- (1) Royal Botanical Garden, Kew, England
 - (2) Indian Botanical Garden, Howrah, Kolkata
 - (3) Pisa, Italy
 - (4) Orto Botanico, Italy
- 131.** Taxonomic key is based on the contrasting characters generally in a pair known as
- (1) Genus
 - (2) Grade
 - (3) Taxon
 - (4) Couplet
- 132.** A place of collection of living plant specimens is
- (1) Botanical Garden
 - (2) Museum
 - (3) Herbarium
 - (4) Key
- 133.** What are the twin character of growth?
- (1) Increase in number and mass
 - (2) Increase in dry and fresh wt.
 - (3) Increase in mass and volume
 - (4) Increase in cell organelle and cell index
- 134.** For unicellular organisms we are not clear about the uses of term like
- (1) Growth and Metabolism
 - (2) Growth and reproduction
 - (3) Metabolism and cellular organisation
 - (4) Consciousness and Metabolism
- 135.** Which of the following organism do not reproduce?
- (1) Worker bees
 - (2) Mules
 - (3) Infertile human being
 - (4) All of the above

- 136.** A true coelom is absent in phylum
(1) Annelida
(2) Echinodermata
(3) Nematoda
(4) Chordata
- 137.** Cell aggregate type body plan found in :-
(1) Cnidaria
(2) Nematoda
(3) Porifera
(4) Arthropoda
- 138.** Deuterostomes are
(1) Echinodermata
(2) Arthropoda
(3) Annelida
(4) Platyhelminthes
- 139.** An opening serving both as mouth and anus is found in
(1) Annelids and Molluscs
(2) Coelenterates and Annelids
(3) Coelenterates and Platyhelminthes
(4) Coelenterates and Aschelminthes
- 140.** Pseudocoelom found in
(1) Mollusca (2) Vertebrata
(3) Arthropoda (4) Aschelminthes
- 141.** A coelom is absent in
(1) Coelenterata
(2) Porifera
(3) Platyhelminthes
(4) All the above
- 142.** If male reproductive part mature early, in bisexual animal the phenomenon is called
(1) Sexual dimorphism
(2) Protandry
(3) Metagenesis
(4) Protogyny
- 143.** The symmetry found in an animal which can be divided into two similar halves by one plane only, into right and left mirror images is
(1) Biradial (2) Spherical
(3) Radial (4) Bilateral
- 144.** Which of the following is the only triploblastic acoelomate phylum?
(1) Porifera
(2) Cnidaria
(3) Ctenophora
(4) Platyhelminthes
- 145.** Fossorial animals live
(1) In burrows
(2) On rocks
(3) On trees
(4) In water bodies
- 146.** A coelom formed by fusion of gut pouches called
(1) Schizocoel
(2) Haemocoel
(3) Enterocoel
(4) Pseudocoel
- 147.** Which of the following are not poikilothermic?
(1) Fish (2) Aves
(3) Reptiles (4) Amphibians
- 148.** Notochord is a rod like structure found on the dorsal side during embryonic development in some animals and is derived from
(1) Mesoderm
(2) Endoderm
(3) Ectoderm
(4) Both Endoderm and Ectoderm
- 149.** In which of the following bilateria, the body cavity is not lined by mesoderm. Instead, the mesoderm is present as scattered pouches inbetween the ectoderm and endoderm?
(1) Echinoderms
(2) Annelids
(3) Platyhelminthes
(4) Aschelminthes
- 150.** In which organisms, the body can be divided into identical left and right halves in only plane in only larval stage?
(1) Arthropoda (2) Mollusc
(3) Annelida (4) Echinoderm

151. Metameric segmentation is present in

- (1) Annelids
- (2) Annelids and Arthropoda
- (3) Roundworms
- (4) Flatworms

152. Examine the diagrammatic sectional view of an animals showing germ layers and coelom.



Choose the correct explanation about it

- (1) Triploblastic and acoelomate
- (2) Triploblastic and coelomate
- (3) Diploblastic and acoelomate
- (4) Diploblastic animal having the thickest layer as mesoglea

153. Match the following list of animals with their level of organization.

Level of organisation	Animal
(A) Organ level	(i) Round worms
(B) Cellular level	(ii) Flat worms
(C) Tissue level	(iii) Sponge
(D) Organ system level	(iv) Cnidarian

Choose the correct match.

- (1) A-(iv), B-(i), C-(ii), D-(iii)
- (2) A-(iv), B-(i), C-(iii), D-(ii)
- (3) A-(ii), B-(iii), C-(iv), D-(i)
- (4) A-(i), B-(iv), C-(iii), D-(ii)

154. Which arrangement is in correct ascending order :-

- (1) Kingdom > Phylum > Order > Class
- (2) Phylum > Class > Order > Kingdom
- (3) Order > Class > Kingdom > Phylum
- (4) Kingdom > Phylum > Class > Order

155. Exoskeleton of CaCO_3 is found in which protozoan :-

- (1) Amoeba
- (2) Foraminiferans
- (3) Radiolarians
- (4) Paramecium

156. Which of the following is characteristic feature of mastigophora?

- (1) Cilia
- (2) Food vacuole
- (3) Flagella
- (4) Pellicle

157. Match the column-I with column-II and select the correct answer :-

Column-I	Column-II
A. Giardia intestinalis	i. Pebrine disease
B. Leishmania tropia	ii. Pyorrhoea
C. Nosema	iii. Diarrhoea
D. Trichomonas tinax	iv. Oriental sore
(1) A-iii, B-ii, C-i, D-iv	
(2) A-ii, B-iv, C-i, D-iii	
(3) A-iii, B-iv, C-i, D-ii	
(4) A-i, B-iv, C-iii, D-ii	

158. Macronucleus of Paramoecium is responsible for :-

- (1) Metabolic activities
- (2) Genetical activities
- (3) Respiratory activities
- (4) All of the above

159. Which organism is considered as of bell animacule?

- (1) Amoeba
- (2) Euglena
- (3) Trypanosoma
- (4) Vorticella

160. In which of the animal dimorphic nucleus is found?

- (1) Amoeba
- (2) Trypanosoma
- (3) Plasmodium
- (4) Paramoecium

161. African sleeping sickness

- (1) Tse Tse fly
- (2) Dragon fly
- (3) Sand fly
- (4) Fruit fly

162. Germ cells of sponges are formed by

- (1) Choanocytes
- (2) Collenocytes
- (3) Chromocytes
- (4) Archaeocytes

- 163.** Common bath sponge is
(1) Sycon (2) Euspongia
(3) Spongilla (4) Poterion
- 164.** Sponges capture food by
(1) Trophocytes (2) Pinacocytes
(3) Choanocytes (4) Silicoblasts
- 165.** In porifera, skeleton forming cells are
(1) Thesocytes (2) Myocytes
(3) Archaeocytes (4) Scleroblast
- 166.** Water current in a sponge is maintained by
(1) Amoebocytes (2) Myocytes
(3) Pinacocytes (4) Choanocytes
- 167.** Choanocytes are
(1) Biflagellate (2) Multiflagellate
(3) Uniflagellate (4) Without flagella
- 168.** Smallest sponge is
(1) Leucosolenia (2) Chalina
(3) Cliona (4) Poterion
- 169.** Study of sponges is known as
(1) Protozoology (2) Parazoology
(3) Helminthology (4) Entomology
- 170.** Sponges are
(1) Mostly marine
(2) Hermaphrodite
(3) Mostly asymmetrical
(4) All of these
- 171.** Pinacoderm layer of body wall in sponges has
(1) Pinacocytes (2) Porocytes
(3) Both (1) and (2) (4) Choanocytes
- 172.** In a sponge, digestion takes place in
(1) Osculum (2) Ostium
(3) Paragastric cavity (4) Food vacuoles
- 173.** Sponges have great regenerating power due to which are cells
(1) Archaeocytes, flagellated
(2) Archaeocytes, totipotent
(3) Scleroblasts, amoeboid
(4) Choanocytes, flagellated
- 174.** Which of the following cannot be considered as function of canal system of sponges?
(1) Respiratory exchange
(2) Formation of gametes
(3) Removal of waste
(4) Food gathering
- 175.** Match the following.
- | Column I | Column II |
|------------------------------------|---------------------------------------|
| (A) Archaeocytes | (i) Store food granules |
| (B) Thesocytes | (ii) Provide food to developing cells |
| (C) Scleroblast | (iii) Totipotent cells |
| (D) Trophocytes | (iv) Secrete spicules |
| (1) A-(iii), B-(i), C-(iv), D-(ii) | |
| (2) A-(i), B-(ii), C-(iii), D-(iv) | |
| (3) A-(i), B-(iii), C-(ii), D-(iv) | |
| (4) A-(iii), B-(iv), C-(i), D-(ii) | |
- 176.** Larva of Hydra is
(1) Planula (2) Hydrula
(3) Absent (4) Dipleura
- 177.** Hydra is
(1) Diploblastic, Bilaterally symmetrical and acoelomate
(2) Diploblastic, Radially symmetrical and acoelomate
(3) Triploblastic, Bilaterally symmetrical and acoelomate
(4) Triploblastic, Radially symmetrical and acoelomate
- 178.** Larva of Aurelia is
(1) Planula (2) Ephyra
(3) Nauplius (4) Trochophore
- 179.** Digestion in Hydra is
(1) Saprozoic
(2) Extra and intracellular
(3) Intracellular only
(4) Extracellular only
- 180.** Mark the cell which first time evolved in Coelenterata
(1) Nerve cell (2) Muscle cell
(3) Gland cell (4) Germ cell

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AIR - 62  RIYA SINGH Classroom	AIR - 65  PRATEEK K. MISHRA Classroom	AIR - 66  VAIBHAV MITTAL Classroom	AIR - 69  VAIBHAV GARG Distance	AIR - 70  ATAN AGARWAL Distance	AIR - 73  SANJEEV REDDY Distance	AIR - 74  SAVI GADEGONE Distance	AIR - 75  PRANJAL PRASAD Distance	AIR - 77  PRAGUN AHUJA Distance	AIR - 78  PARVA N. LADANI Distance	AIR - 79  PULKIT GUPTA Classroom	AIR - 81  V.N.T.V. BELLAMKONDA Distance	AIR - 83  PRABHJOT SINGH Classroom	AIR - 84  RUTUDHWAJ SAVALIYA Classroom	AIR - 90  EISHWARI AGGARWAL Distance
AIR - 91  SHUBHAM MARODIA Distance	AIR - 95  KSHITIJ AGARWAL Classroom	81230* Classroom 50267 Distance 30963										AIR - 96  JITENDRA YADAV Classroom	AIR - 100  VIDUSHI VARSHEY Classroom	

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